DTMT 2026
Subject Round: Algebra Solutions

Q1. Sitar's cortisol level after x hours of watching reels is modeled by $c(x) = -\frac{1}{2}x^2 + 6x + 7$. How many hours should Sitar scroll to maximize his cortisol levels?

Solution: Sitar's cortisol function is

$$c(x) = -\frac{1}{2}x^2 + 6x + 7,$$

which is a downward-opening parabola. The maximum occurs at the vertex.

For a quadratic $ax^2 + bx + c$, the x-coordinate of the vertex is

$$x = -\frac{b}{2a}.$$

Here, $a = -\frac{1}{2}$ and $b = 6$, so

$$x = -\frac{6}{2(-\frac{1}{2})} = 6.$$

Thus Sitar should scroll for $\boxed{6}$ hours.

Q2. Hansel and Gretel discover a house made entirely of candy! On the first night, they act cautiously, as to not get caught by the witch, and collectively (meaning both of them together) sneak only 5 pieces of candy. As they gain confidence, they increase their loot by 3 pieces every single night. However, every 5th night (that is, nights 5, 10, 15, ...), they become nervous and sneak 4 fewer pieces than they otherwise would have that night. How many pieces of candy will they sneak on the 20th night?

Solution:

$$a_1 = 5$$

Each night, the amount increases by 3, except that on every 5th night they take 4 fewer than they would have taken that night.

So the amounts are:

$$a_2 = 8, \quad a_3 = 11, \quad a_4 = 14$$

On the 5th night, they would have taken 17, but they take 4 fewer:

$$a_5 = 17 - 4 = 13$$

Then the pattern continues from the previous night's amount:

$$a_6 = 16,\ a_7 = 19,\ a_8 = 22,\ a_9 = 25,\ a_{10} = 28 - 4 = 24$$

$$a_{11} = 27,\ a_{12} = 30,\ a_{13} = 33,\ a_{14} = 36,\ a_{15} = 39 - 4 = 35$$

$$a_{16} = 38,\ a_{17} = 41,\ a_{18} = 44,\ a_{19} = 47,\ a_{20} = 50 - 4 = 46$$

Therefore, the number of pieces of candy they sneak on the 20th night is

$$\boxed{46}.$$

Q3. The *D.licious* cooking class says the grapes in the cafeteria are basically water. Fresh grapes are 90% water by weight, while dried raisins are 20% water by weight. After drying, 2 kilograms of raisins are accidentally discarded before packaging. If the final packaged raisins weigh 16 kilograms, how many kilograms of fresh grapes were originally used?

Solution: Since 2 kg were discarded and 16 kg were packaged, the original mass of raisins was

$$16 + 2 = 18 \text{ kg}.$$

Raisins are 20% water, so they are 80% solids. Thus the solids weigh

$$0.80 \cdot 18 = 14.4 \text{ kg}.$$

Fresh grapes are 90% water, so they are 10% solids. Let m be the mass of fresh grapes. Then

$$0.10m = 14.4 \quad \Rightarrow \quad m = \boxed{144}.$$

Q4. The graph of $y = x^2$ is translated h units to the right and k units down, where h and k are integers. The translated parabola intersects the x-axis at $(1, 0)$ and $(7, 0)$. Compute $k - h$.

Solution: After translating right h and down k, the equation is

$$y = (x - h)^2 - k.$$

The x-intercepts $(1, 0)$ and $(7, 0)$ satisfy

$$(1 - h)^2 - k = 0, \qquad (7 - h)^2 - k = 0.$$

So $(1 - h)^2 = (7 - h)^2$, which implies

$$1 - h = -(7 - h) \Rightarrow 1 - h = h - 7 \Rightarrow 2h = 8 \Rightarrow h = 4.$$

Then

$$k = (1 - 4)^2 = 9.$$

Thus $k - h = 9 - 4 = \boxed{5}$.

Q5. In the library, *d.ctionary* inventories the *odd-numbered* bookshelves and a friend, *d.ssertations* inventories the *even-numbered* bookshelves. Shelf 1 has 6 books and shelf 2 has 9 books. The numbers of books on the odd-numbered shelves form an arithmetic progression, and the numbers of books on the even-numbered shelves form an arithmetic progression. *d.ctionary* counts a total of 576 books on the odd-numbered shelves, and the friend counts a total of 744 books on the even-numbered shelves. If the last shelf has 84 books, how many shelves are there in total?

Solution: Let there be m odd-numbered shelves and n even-numbered shelves.

Odd shelves form an arithmetic progression with first term 6 and sum 576:

$$\frac{m}{2}\big(2 \cdot 6 + (m-1)d_1\big) = 576.$$

Even shelves form an arithmetic progression with first term 9 and sum 744:

$$\frac{n}{2}\big(2 \cdot 9 + (n-1)d_2\big) = 744.$$

The last shelf has 84 books. If the total number of shelves is even, the last shelf is even-numbered, so

$$9 + (n-1)d_2 = 84 \Rightarrow (n-1)d_2 = 75.$$

Then the even-shelf sum becomes

$$\frac{n}{2}(18 + 75) = \frac{93n}{2} = 744 \Rightarrow 93n = 1488 \Rightarrow n = 16.$$

So $m = n = 16$ and the total number of shelves is

$$m + n = 32.$$

If the total number of shelves were odd, the last shelf would be odd-numbered, giving

$$6 + (m-1)d_1 = 84 \Rightarrow (m-1)d_1 = 78,$$

and then the odd-shelf sum would be

$$\frac{m}{2}(12 + 78) = 45m = 576,$$

which is not an integer m. So the total cannot be odd.

Therefore, there are $\boxed{32}$ shelves.

Q6. The Tortoise and the Hare compete in a race on a straight track of length 156 meters. The Hare runs at a constant integer speed v_h (in meters per minute), and the Tortoise runs at a constant integer speed v_t. The Hare runs for 16 minutes, sleeps until minute 120, then resumes running at his original pace, v_h.

Surprisingly, both the Tortoise and the Hare cross the finish line at the same time. Find the time, in minutes, after the start of the race, when the Tortoise overtakes the sleeping Hare.

Solution: Let the finish time be T minutes. Then the Tortoise travels

$$156 = v_t T.$$

The Hare runs for 16 minutes, sleeps until minute 120, then runs for $T - 120$ more minutes, so his total running time is

$$16 + (T - 120) = T - 104,$$

and

$$156 = v_h(T - 104).$$

Since v_t and v_h are integers, T and $T - 104$ must both divide 156. Also $T > 120$.

The divisors of 156 greater than 120 give $T = 156$. Then

$$T - 104 = 52,$$

which also divides 156, so this is consistent. Thus

$$v_t = \frac{156}{156} = 1, \qquad v_h = \frac{156}{52} = 3.$$

The Hare sleeps starting at minute 16 at distance

$$3 \cdot 16 = 48.$$

The Tortoise reaches 48 meters at time 48 since $v_t = 1$. Therefore, the Tortoise overtakes the sleeping Hare at $\boxed{48}$ minutes.

Q7. Three neon signs hang outside the d.tech bathrooms, labeled A, B, and C. Each sign blinks at a constant integer interval of seconds, t_A, t_B, and t_C, such that $t_A < t_B < t_C$.

The signs are turned on simultaneously at $t = 0$. Dylan, a keen observer, records the following pattern:

- Signs A and B blink together precisely every 180 seconds.
- Signs B and C blink together precisely every 360 seconds.

Find the sum of all possible distinct integer values for t_C.

Solution: The conditions mean

$$\operatorname{lcm}(t_A, t_B) = 180, \qquad \operatorname{lcm}(t_B, t_C) = 360.$$

Factor

$$180 = 2^2 \cdot 3^2 \cdot 5, \qquad 360 = 2^3 \cdot 3^2 \cdot 5.$$

Since t_B divides 180, it has at most 2^2. Therefore, to make $\operatorname{lcm}(t_B, t_C)$ have a factor 2^3, we must have $2^3 \mid t_C$, so

$$t_C \in \{8 \cdot 3^v \cdot 5^w : 0 \le v \le 2,\ 0 \le w \le 1\} = \{8, 24, 40, 72, 120, 360\}.$$

Now check each possibility using $t_B < t_C$ and the lcm conditions.

(1) $t_C = 8$ or $t_C = 24$: To have $\operatorname{lcm}(t_B, t_C) = 360$, t_B must supply 3^2 and 5, so $45 \mid t_B$. Then $t_B \ge 45 > t_C$, contradicting $t_B < t_C$. So these are impossible.

(2) $t_C = 40$: We need $\operatorname{lcm}(t_B, 40) = 360$, so $9 \mid t_B$. Take $t_B = 36$, then $\operatorname{lcm}(36, 40) = 360$. Also $\operatorname{lcm}(5, 36) = 180$ with $5 < t_B$, so $(t_A, t_B, t_C) = (5, 36, 40)$ works. Thus 40 is possible.

(3) $t_C = 72$: We need $\operatorname{lcm}(t_B, 72) = 360$, so $5 \mid t_B$. Take $t_B = 45$, then $\operatorname{lcm}(45, 72) = 360$. Also $\operatorname{lcm}(4, 45) = 180$ with $4 < t_B$, so $(t_A, t_B, t_C) = (4, 45, 72)$ works. Thus 72 is possible.

(4) $t_C = 120$: We need $\operatorname{lcm}(t_B, 120) = 360$, so $9 \mid t_B$. Take $t_B = 45$, then $\operatorname{lcm}(45, 120) = 360$. Also $\operatorname{lcm}(4, 45) = 180$ with $4 < t_B$, so $(t_A, t_B, t_C) = (4, 45, 120)$ works. Thus 120 is possible.

(5) $t_C = 360$: Take $t_B = 180$, then $\operatorname{lcm}(180, 360) = 360$, and choose $t_A = 1$ so $\operatorname{lcm}(1, 180) = 180$. Thus 360 is possible.

Therefore the possible values of t_C are

$$40,\ 72,\ 120,\ 360,$$

and their sum is

$$40 + 72 + 120 + 360 = \boxed{592}.$$

Q8. A sequence of numbers $\{a_n\}$ is defined by $a_1 = 1$, $a_2 = 1$, and for all $n \geq 1$:

$$a_{n+2} = \frac{a_{n+1}^2 + 2}{a_n}$$

Find the value of:

$$\frac{a_{15} + a_{13}}{a_{14}}$$

Solution: We have

$$a_{n+2}a_n = a_{n+1}^2 + 2.$$

Shifting the index by -1 gives

$$a_{n+1}a_{n-1} = a_n^2 + 2.$$

So for every $n \geq 2$,

$$a_n^2 + 2 = a_{n+1}a_{n-1}.$$

Claim: for all $n \geq 2$,

$$a_{n+1} = 4a_n - a_{n-1}.$$

Check $n = 2$: $a_3 = \frac{1^2+2}{1} = 3 = 4 \cdot 1 - 1$.

Assume the relation holds for a_{n+1}, meaning $a_{n+1} = 4a_n - a_{n-1}$. Then

$$a_{n+2} = \frac{a_{n+1}^2 + 2}{a_n} = \frac{(4a_n - a_{n-1})^2 + 2}{a_n} = \frac{16a_n^2 - 8a_n a_{n-1} + a_{n-1}^2 + 2}{a_n}.$$

Use $a_{n-1}^2 + 2 = a_n a_{n-2}$ from the shifted identity:

$$a_{n+2} = 16a_n - 8a_{n-1} + a_{n-2}.$$

Also shift the claimed relation once: $a_n = 4a_{n-1} - a_{n-2}$, so $a_{n-2} = 4a_{n-1} - a_n$. Substitute:

$$a_{n+2} = 16a_n - 8a_{n-1} + (4a_{n-1} - a_n) = 15a_n - 4a_{n-1}.$$

But

$$4a_{n+1} - a_n = 4(4a_n - a_{n-1}) - a_n = 15a_n - 4a_{n-1},$$

so

$$a_{n+2} = 4a_{n+1} - a_n.$$

This is exactly the same form with the index advanced, so the claim holds for all $n \geq 2$.

Now apply it at $n = 14$:

$$a_{15} = 4a_{14} - a_{13}.$$

Then

$$\frac{a_{15} + a_{13}}{a_{14}} = \frac{(4a_{14} - a_{13}) + a_{13}}{a_{14}} = \frac{4a_{14}}{a_{14}} = \boxed{4}.$$

Q9. At the 2025 Design Tech Math Tournament, the top three individual scores were distinct positive integers denoted by d, t, and m. These scores satisfied this system of equations:

$$d + t + m = 23 \tag{1}$$

$$d^2 + t^2 + m^2 = 227 \tag{2}$$

$$\frac{1}{d+t} + \frac{1}{t+m} + \frac{1}{m+d} = \frac{17}{80} \tag{3}$$

Find the product of the top three scores, $d \cdot t \cdot m$.

Solution: Let

$$s_1 = d + t + m, \qquad s_2 = dt + tm + md, \qquad s_3 = dtm.$$

From (1) we have $s_1 = 23$. Using

$$(d + t + m)^2 = d^2 + t^2 + m^2 + 2(dt + tm + md),$$

and (2), we get

$$23^2 = 227 + 2s_2 \Rightarrow 529 = 227 + 2s_2 \Rightarrow s_2 = 151.$$

Let

$$A = d + t, \quad B = t + m, \quad C = m + d.$$

Then

$$A + B + C = 2(d + t + m) = 46,$$

and

$$ABC = (d + t)(t + m)(m + d) = (d + t + m)(dt + tm + md) - dtm = s_1 s_2 - s_3.$$

Also,

$$AB = (d + t)(t + m) = dt + dm + t^2 + tm,$$

and similarly for BC and CA. Adding gives

$$AB + BC + CA = (d^2 + t^2 + m^2) + 3(dt + tm + md) = 227 + 3 \cdot 151 = 680.$$

Therefore,

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} = \frac{AB + BC + CA}{ABC} = \frac{680}{s_1 s_2 - s_3}.$$

Using (3),

$$\frac{680}{23 \cdot 151 - s_3} = \frac{17}{80}.$$

So

$$80 \cdot 680 = 17(3473 - s_3) \Rightarrow 54400 = 59041 - 17s_3 \Rightarrow 17s_3 = 4641 \Rightarrow s_3 = \boxed{273}.$$

Q10. A standard diagram of the trefoil knot has three crossings. In a *right-handed* trefoil, all crossings are positive; in its mirror image (the left-handed trefoil), all crossings are negative.

At each step i, we encounter a crossing in the diagram. Define

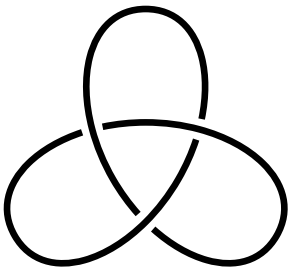
$$s_i = \begin{cases} +1, & \text{if the strand passes over at the } i\text{-th encountered crossing,} \\ -1, & \text{if the strand passes under at the } i\text{-th encountered crossing.} \end{cases}$$

Let

$$T = -\sum_{i=1}^{2026} i\, s_i\, (-1)^{i-1}.$$

For the right-handed trefoil diagram shown below, compute $T \pmod{1000}$.

The Trefoil Knot



Solution: In the standard right-handed trefoil diagram, as you traverse the knot you encounter crossings in an alternating *over, under, over, under, ...* pattern, so

$$s_i = (-1)^{i-1}.$$

Then each summand simplifies:

$$i\, s_i\, (-1)^{i-1} = i \cdot (-1)^{i-1} \cdot (-1)^{i-1} = i,$$

so

$$T = -\sum_{i=1}^{2026} i = -\frac{2026 \cdot 2027}{2}.$$

Compute:

$$\frac{2026 \cdot 2027}{2} = 1013 \cdot 2027 = 2{,}053{,}351.$$

Thus

$$T \equiv -2{,}053{,}351 \equiv -351 \equiv 649 \pmod{1000}.$$

Therefore, $\boxed{649}$.